



PROBLEM-3

Compute difference

$d_i = \text{favourite algo. error} - \text{Decision Tree error}$

fold	favourite	Decision Tree	d_i
1	8.89	9.30	-0.41
2	9.52	9.46	0.04
3	8.13	9.12	-0.99
4	9.48	9.13	0.35
5	10.12	9.98	0.14
6	10.23	11.01	-0.78
7	8.56	9.02	-0.46
8	9.12	8.56	0.56
9	9.23	9.23	0.00
10	9.11	9.08	0.03

Mean difference

$$\bar{d} = \frac{\sum d_i}{n} = \frac{-1.52}{10} = -0.152$$

Negative value indicates that favourite algo. has slightly lower error.

S.D of difference

$$s_d = 0.494$$



→ Compute t-statistic

$$t = \frac{\bar{a}}{s_d / \sqrt{n}}$$

$$t = \frac{-0.152}{0.494 / \sqrt{10}} = \frac{-0.152}{0.156} = -0.97$$

→ Degree of freedom
 $df = n - 1 = 9$

for 95% confidence, critical value
 $t_{critical} \approx 2.262$

$$|t| = 0.97 < 2.262$$

Result is not statistically significant.